Objet Fwd: From the g-2 Anomaly to Muonic Propulsion: Toward a New Rocket Paradigm
À : [null <hossenfelder@uni-muenchen.de>, null <sabine.hossenfelder@gmail.com>]
De Robert Dosseh-Kpotogbey <rdossehkpotogbey@gmail.com>
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Date : lun., mars 23, 2026 à 17:30

Abstract

This paper develops a theoretical framework connecting the muon magnetic anomaly (*g*-2) to a novel concept of particle-based space propulsion. We attribute part of the muon's anomalous magnetic moment to an internal geometric compactness of massive leptons, characterized by an effective radius *r_ℓ = κ/m_ℓ* inversely proportional to lepton mass. This structure produces a Casimir energy amplification between the muon and the electron of order $(207)^5 \approx 4 \times 10^{11}$, and between the tauon and the electron of order $(3477)^5 \approx 5 \times 10^{17}$.

Within a generalized *n*-dimensional dynamical framework (*n* ≈ 5), consistent with the quintic mass dependence of anomalous magnetic moment contributions, we derive a velocity law for particles produced in lepton-lepton collisions: **$vf^2 = v_2^2 \cdot (m_\ell/m_0)^{(n-2)}$**. Applied to muon-muon collisions, this yields an amplification factor of $(207)^3 \approx 8.9 \times 10^6$ and a maximum ejection velocity of ~20,700 km/s, corresponding to a specific impulse approximately 6,000 times that of state-of-the-art chemical rocket engines. Applied to tauon-tauon collisions, the amplification factor reaches $(3477)^3 \approx 4.2 \times 10^{10}$, with ejection velocities potentially exceeding 10^{12} m/s within the generalized formalism.

Two nested propulsive regimes emerge naturally from the three-generation leptonic hierarchy: a **muonic regime** targeting interplanetary missions, and a **tauonic regime** targeting interstellar missions, with mathematical continuity ensured at the domain boundary. Recent experimental advances in compact muon production — including laser-driven sources (CAEP/CAS, *Nature Physics* 2025), radiofrequency acceleration (J-PARC), and ionization cooling (MICE) — are discussed as enabling technologies for a prospective muonic thruster architecture. The broader implications for high-energy astrophysical environments, including black hole jets and neutrino production mechanisms, are outlined.

Keywords: muon g-2 anomaly, leptonic compactness, Casimir effect, generalized n-dimensional dynamics, muonic propulsion, tauonic propulsion, specific impulse, interstellar propulsion.

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De : **Robert Dosseh-Kpotogbey** <rdossehkpotogbey@gmail.com>
Date: lun. 23 mars 2026, 17:21
Subject: Fwd: From the g-2 Anomaly to Muonic Propulsion: Toward a New Rocket Paradigm
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De : **Robert Dosseh-Kpotogbey** <rdossehkpotogbey@gmail.com>
Date: lun. 23 mars 2026, 14:58
Subject: From the g-2 Anomaly to Muonic Propulsion: Toward a New Rocket Paradigm
To: Origin of black hole jets, neutrinos, dark matter and dark energy <origin-of-black-hole-jets-neutrinos-dark-matter-and-dark-energy@googlegroups.com>

From the g-2 Anomaly to Muonic Propulsion: Toward a New Rocket Paradigm

I. The Muon Magnetic Anomaly as a Signal of New Physics

For several decades, the muon magnetic anomaly — commonly denoted *g-2* — has stood as one of the most persistent puzzles in fundamental physics. The anomalous magnetic moment *a_μ* of the muon displays a measurable deviation from the predictions of the Standard Model, a discrepancy that, though small in absolute value, exceeds by several standard deviations what quantum field theory can account for through the known QED, electroweak, and hadronic contributions alone.

Our previous work proposed an explanatory path grounded not in the existence of exotic new particles, but in a geometric internal structure intrinsic to massive leptons. Within this framework, the total magnetic anomaly of a lepton is written:

$a_\ell = a_{QED} + a_{EW+had}(m_\ell^5) + a_{str}(m_\ell, G_\ell)$

where the term *a_str* encodes the effects of an internal geometric compactness, characterized by an effective radius *$r_\ell = \kappa/m_\ell$*, inversely proportional to mass. The more massive the lepton, the more confined its internal structure, and the more amplified its interaction with the quantum vacuum — notably through the Casimir effect.

The Casimir energy amplification ratio between the muon and the electron is striking:

$E_{Casimir,\mu} / E_{Casimir,e} \approx (207)^5 \approx 4 \times 10^{11}$

This colossal factor, arising solely from the mass ratio *$m_\mu/m_e \approx 207$*, means that the muon interacts with vacuum fluctuations in a fundamentally different manner than the electron. Beyond its interest for fundamental physics, this observation leads us to formulate a technological hypothesis: if this structural difference translates into a distinct capacity to acquire and transfer kinetic energy during collisions, then **muon-muon collisions** — rather than electron-electron collisions — could generate particles endowed with radically superior propulsive properties.

II. From Muonic Collisions to the Production of Propulsive Particles

2.1 Why Muon-Muon Collisions?

Current high-energy frontier colliders primarily use protons (LHC at CERN) or electrons (LEP, ILC in planning). Protons, as hadrons, are composite objects: their collisions generate a complex QCD background that is difficult to control. Electrons and positrons are point-like leptons whose collisions are clean, but their low mass severely limits the energy available in the center-of-mass frame.

The muon, by contrast, occupies a unique position. With a mass 207 times greater than that of the electron, it is:

- **Far less subject to synchrotron radiation** than the electron (energy loss by radiation scales as *$1/m^4$*, giving a factor of *$(207)^4 \approx 1.8 \times 10^9$* in favor of the muon),
- **Capable of reaching much higher collision energies** in a circular machine of comparable size,
- **A point-like lepton**, so its collisions remain clean, without parasitic hadronic background.

A high-energy muon-muon collider would reach center-of-mass energies of several TeV in a ring only a few kilometers in circumference, whereas the LHC (27 km circumference) barely reaches 14 TeV with protons, of which only a fraction of the energy is available for the collision of elementary quarks.

Within our extended theoretical framework, the generalized dynamics is written:

$((m_0c^2/\sqrt{(1 - v^n/c^n)})^2)^{(n/2)} - ((m_0vc/\sqrt{(1 - v^n/c^n)})^2)^{(n/2)} = (m_0^2c^4)^{(n/2)}$

with *$n \approx 5$* as the characteristic value suggested by the quintic mass dependence of contributions to the magnetic anomaly. This framework introduces generalized units *$[C_n] = [v_n] = m^{(2/n)}/s$*, and the relation to ordinary velocity:

$v_n \cdot dt_0^{(n-2)} = v_2^2$

with the key substitution *$q = m_\mu/m_0$* linking the two regimes. When particles are produced in muon-muon collisions within this extended framework, their final velocity *vf* follows:

$vf^2 = v_2^2 \cdot (m_\mu/m_0)^{(n-2)}$

For *$n \approx 5$* and *$v_2 \in]0, 25{,}000$ km/h]*, the amplification factor is:

$(m_\mu/m_0)^3 = (207)^3 \approx 8.9 \times 10^6$

This extraordinary amplification factor means that particles produced in muon-muon collisions could, in principle, reach speeds considerably higher than those generated by electron-electron collisions at comparable collision energy.

III. Application to Space Propulsion: Replacing Chemical Propellants

3.1 The Limits of Current Propellants

The most powerful contemporary rockets rely on two families of chemical propellants:

First family — kerosene / LOX:
- **Fuel**: RP-1 (Rocket Propellant-1), a high-purity refined kerosene
- **Oxidizer**: Liquid oxygen (LOX)
- Used notably by SpaceX's Merlin engine, specific impulse of approximately **310 to 350 seconds**

in vacuum

Second family — methane / LOX:
- **Fuel**: Liquid methane (CH_4), SpaceX's Raptor propellant
- **Oxidizer**: Liquid oxygen (LOX)
- Raptor engine specific impulse: **363 seconds** in vacuum, with thrust exceeding 2 MN

Both systems, however high-performing, are fundamentally constrained by the **Tsiolkovsky equation**:

$\Delta v = Isp \cdot g_0 \cdot \ln(m_0/mf)$

The specific impulse *Isp* is directly linked to the exhaust velocity of gases: *$Isp = ve/g_0$*. For chemical propellants, this exhaust velocity is limited to a few km/s, determined by the enthalpy of the chemical reaction. No chemical propellant can physically exceed 5,000 m/s exhaust velocity — an absolute thermodynamic constraint.

3.2 The Concept of Muonic Particle Propulsion

Our central hypothesis is as follows: **particles produced in high-energy muon-muon collisions, accelerated and directed through a nozzle divergent, could substitute for the combustion gases of chemical propellants**, providing exhaust velocities incomparably beyond those of chemical reactions.

From the velocity relation:

$vf^2 = v_2^2 \cdot (m_\mu/m_0)^{(n-2)}$

with *$n \approx 5$*, *$v_2 \leq 25{,}000$ km/h $\approx 6{,}944$ m/s*:

$vf \leq \sqrt{(6944^2 \times (207)^3)} \approx \sqrt{(4.82 \times 10^7 \times 8.9 \times 10^6)} \approx \sqrt{(4.29 \times 10^{14})} \approx 2.07 \times 10^7$ m/s

That is, approximately **20,700 km/s**. Such an exhaust velocity would represent a specific impulse on the order of **2×10^6 seconds** — **6,000 times** the specific impulse of a Raptor engine.

This performance would open the door to orbital, interplanetary, and potentially interstellar maneuvers under a completely rewritten Tsiolkovsky equation: a spacecraft equipped with such a propulsion system would need only a tiny mass of muonic propellant to achieve considerable delta-v values.

IV. Technological Feasibility and State of the Art in Muon Production

4.1 Laser-Driven Muon Production: A Revolution Underway

The feasibility of this concept rests first on the ability to produce and accelerate muons in useful quantities. Recent decisive advances have been made on this front.

Researchers from the China Academy of Engineering Physics (CAEP), Guangdong Laboratory, and the Chinese Academy of Sciences (CAS) have developed a muon production method using **ultra-short high-intensity laser pulses**, published in *Nature Physics* (2025). The device relies on laser wakefield acceleration (LWFA) of an electron beam to GeV energies, directed through a lead converter target to generate muons. The measured yield reaches **0.01 muon per incident electron**, with a theoretical estimate of **10^7 muons per laser shot**.

In parallel, a team at the Japan Proton Accelerator Research Complex (J-PARC) achieved the **first

stable acceleration of positive muons from thermal energy to 100 keV** using a radiofrequency cavity. The phase-space reduction achieved — factors of 200 (horizontal) and 410 (vertical) — marks a decisive step toward compact and efficient muonic accelerators.

Finally, the MICE (Muon Ionization Cooling Experiment) collaboration demonstrated **reduction of the transverse emittance of muon beams** through ionization cooling in lithium hydride and liquid hydrogen absorbers, enabling the production of bright beams required for high-luminosity collisions.

4.2 Architecture of a Muonic Propulsion System

Based on these advances, a conceptual muonic thruster architecture could be structured as follows:

1. **Compact muon source**: High-power CPA (Chirped Pulse Amplification) laser generating a GeV electron beam via LWFA, striking a solid converter to produce muons.
2. **Beam cooling and focusing**: Via ionization cooling in an appropriate absorber, reducing the muon beam emittance.
3. **Onboard annular collider**: Miniaturized muon-muon collision ring at high radiofrequency voltage, enabling repeated collisions at controlled energy within the range *$v_2 \in]0, 25{,}000$ km/h]*.
4. **Production and extraction chamber**: Magnetic enclosure guiding produced particles from muon-muon collisions toward the ejection nozzle.
5. **Electromagnetic nozzle**: Magnetic divergent converting the momentum of charged particles into vectorized thrust along the vehicle's longitudinal axis.

V. Conclusion of the Muonic Regime

The muon magnetic anomaly *g-2*, far from being a mere academic curiosity, appears in our theoretical model as the signal of a **geometric internal physics** intrinsic to massive leptons — amplified by a factor *$(m_\mu/m_0)^5 \approx 1.8 \times 10^9$* relative to the electron.

The generalized velocity relation **$vf^2 = v_2^2 \cdot (m_\mu/m_0)^{(n-2)}$** with *$n \approx 5$* predicts that particles from muon-muon collisions could reach exhaust velocities exceeding those of all current chemical propellants — RP-1/LOX or liquid methane/LOX — by nearly seven orders of magnitude in specific impulse.

VI. The Tau Lepton: Third-Generation Lepton and Its Remarkable Properties

Having established the propulsive advantage of muon-muon collisions over electron-electron collisions, we now turn to the analogous particle of the **third lepton generation**: the **tauon** (τ^-), discovered in 1976 by Martin Lewis Perl. Its fundamental properties are:

- **Mass**: *$m_\tau = 1{,}777$ MeV/c^2*, i.e., *3.17×10^{-27} kg*
- **Electric charge**: *$-e$* (identical to the electron and muon)
- **Spin**: *1/2* (fermion)
- **Lifetime**: *$\approx 2.9 \times 10^{-13}$ s*
- **Antiparticle**: the antitauon (τ^+)

The tauon-to-electron mass ratio is:

$m_\tau / m_0 \approx 1777 / 0.511 \approx 3{,}477$

approximately **16.8 times** the muon-to-electron ratio (207). The tauon can also form exotic atoms: *tauonium* (τ^+e^-) and *true tauonium* ($\tau^+\tau^-$), the latter being difficult to observe due to the very short

lifetime of the tauon at the low energies required for the formation of such a bound state.

VII. Casimir Amplification in the Tauonic Regime

In our geometric model, the effective radius of the tauon is:

$r_\tau = \kappa / m_\tau \approx r_0 / 3{,}477$

The associated Casimir energy follows:

$E_{Casimir,\tau} \propto -m_\tau^5$

The amplification ratio relative to the electron:

$E_{Casimir,\tau} / E_{Casimir,e} \approx (3{,}477)^5 \approx 5.1 \times 10^{17}$

versus $*(207)^5 \approx 4 \times 10^{11}*$ for the muon. The tauon amplifies interactions with the quantum vacuum by a factor **approximately 1.3×10^6 times greater** than the muon — this colossal amplification propagates directly into the velocity law for particles produced by tauon-tauon collisions.

VIII. Velocity Law in the Generalized n-Dimensional Framework

8.1 Framework Recap

Our theoretical framework rests on generalized units:

$[C_n] = [v_n] = m^{(2/n)} / s$

with the relation between generalized and ordinary velocity:

$v_n^n \cdot dt_0^{(n-2)} = v_2^2$

The substitution parameter $*q = m_\tau / m_0*$ introduces a scale transformation:

$v_n \cdot (q \cdot dt_0)^{(n-2)} = v_2^2 \cdot q^{(n-2)}$

The velocity law for particles produced in tauon-tauon collisions, for $*n \approx 5*$, is:

$vf^2 = v_2^2 \cdot (m_\tau / m_0)^{(n-2)} = v_2^2 \cdot (3{,}477)^3$

with:

$(3{,}477)^3 \approx 4.20 \times 10^{10}$

This formalism is applied **strictly in its own form**, without introducing the Lorentz factor $*1/\sqrt{(1 - v^2/c^2)}*$ or any relativistic effect in the sense of standard special relativity.

8.2 Domain of Application: $v_2 \in]25{,}000$ km/h, 20,700 km/s]

The lower bound $*v_2 = 25{,}000$ km/h $\approx 6{,}944$ m/s* corresponds to the upper limit of the muonic domain previously treated. The upper bound $*v_2 = 20{,}700$ km/s $= 2.07 \times 10^7$ m/s* corresponds precisely to the maximum ejection velocity predicted in the muonic regime, which becomes the

precisely to the maximum ejection velocity predicted in the muonic regime, which becomes the **input value** of the tauonic regime — ensuring physical continuity between the two domains.

IX. Ejection Velocity Calculations in the Tauonic Regime

9.1 At the lower bound: v_2 = 25,000 km/h ≈ 6,944 m/s

$vf^2 = (6{,}944)^2 \times 4.20 \times 10^{10} \approx 2.02 \times 10^{18}$ m²/s²
$vf \approx 1.42 \times 10^9$ m/s ≈ 1,420,000 km/s

9.2 At the median value: v_2 = 500,000 km/h ≈ 1.389×10^5 m/s

$vf^2 = (1.389 \times 10^5)^2 \times 4.20 \times 10^{10} \approx 8.10 \times 10^{20}$ m²/s²
$vf \approx 9.0 \times 10^{10}$ m/s ≈ 90,000,000 km/s

9.3 At the upper bound: v_2 = 20,700 km/s = 2.07×10^7 m/s

$vf^2 = (2.07 \times 10^7)^2 \times 4.20 \times 10^{10} \approx 1.80 \times 10^{25}$ m²/s²
$vf \approx 1.34 \times 10^{12}$ m/s ≈ 1.34×10^9 km/s

These values, computed within the generalized n-dimensional formalism, are not bounded by the speed of light *c* of standard special relativity, but by the generalized speed limit *C_n* proper to the n-dimensional velocity space defined by the author's theoretical framework.

X. Comparative Table: Muon / Tauon Regimes

Parameter	Muonic Regime	Tauonic Regime
Collision particle	μ^- / μ^+	τ^- / τ^+
Mass ratio m/m_e	207	3,477
Factor $(m/m_e)^3$	~8.9×10^6	~4.2×10^{10}
Casimir amplification	~4×10^{11}	~5×10^{17}
v_2 domain	]0 ; 25,000 km/h]	]25,000 km/h ; 20,700 km/s]
vf at lower bound	~0	~1.42×10^9 m/s
vf at upper bound	~2.07×10^7 m/s	~1.34×10^{12} m/s
Propulsive application	Interplanetary missions	Interstellar missions
Propellants replaced	RP-1/LOX, CH_4/LOX	Muonic propulsion itself

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XI. Physical Interpretation and Continuity Between the Two Regimes

The two-domain nested structure reflects a **natural hierarchy in leptonic physics**. In the **muonic regime** *$v_2 \in$]0, 25,000 km/h]*, particles produced by $\mu^-\mu^+$ collisions replace conventional chemical propellants and enable missions throughout the inner and outer solar system with unprecedented specific impulses. In the **tauonic regime** *$v_2 \in$]25,000 km/h, 20,700 km/s]*, particles from $\tau^-\tau^+$ collisions take over exactly where the muonic regime reaches its own ceiling. The entry velocity of the tauonic domain coincides exactly with the maximum output velocity of the muonic domain, ensuring **mathematical and physical continuity** between the two regimes.

This two-level hierarchy — muon for the interplanetary domain, tauon for the interstellar domain — is a direct consequence of the **three-generation structure of charged leptons** in the Standard Model,

and suggests that the leptonic mass hierarchy may not be an arbitrary curiosity of fundamental physics, but could encode a **natural scale of propulsive capabilities** accessible to a technologically advanced civilization.

This text constitutes a prospective extension of the n-dimensional formal framework developed by the author. All calculations are conducted within this proper formalism, independently of standard special relativity and without recourse to the Lorentz factor.

Posted to: BHI – Black Hole Initiative Google Group (cfa.harvard.edu) & Origin of Black Hole Jets, Neutrinos, Dark Matter and Dark Energy Group

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Dear colleagues,

I would like to share with this community a theoretical proposal that connects two areas of active interest here — anomalous lepton magnetic moments and high-energy leptonic physics — with a speculative but mathematically grounded application to propulsion, developed within a generalized n-dimensional dynamical framework.

Core idea. Starting from the well-known g-2 anomaly of the muon, our model attributes part of the discrepancy with the Standard Model to a geometric internal compactness of massive leptons, characterized by an effective radius *r_ℓ = κ/m_ℓ*. This leads to a Casimir energy amplification between muon and electron of order *$(207)^5 \approx 4 \times 10^{11}$*, and between tauon and electron of order *$(3477)^5 \approx 5 \times 10^{17}$*.

The velocity law. Within a generalized n-dimensional framework (n ≈ 5, consistent with the quintic mass dependence of anomalous magnetic moment contributions), particles produced in lepton-lepton collisions follow:

$vf^2 = v_2^2 \cdot (m_\ell / m_0)^{(n-2)}$

This formalism uses generalized units *$[C_n] = m^{(2/n)}/s$* and is applied strictly without the standard Lorentz factor.

Two nested propulsive regimes emerge naturally:

— **Muonic regime**, $v_2 \in]0, 25{,}000$ km/h]: amplification factor $(207)^3 \approx 8.9 \times 10^6$, maximum ejection velocity ~20,700 km/s. Replaces RP-1/LOX and CH_4/LOX chemical propellants for interplanetary missions.

— **Tauonic regime**, $v_2 \in]25{,}000$ km/h, 20,700 km/s]: amplification factor $(3477)^3 \approx 4.2 \times 10^{10}$, maximum ejection velocity ~1.34×10^{12} m/s. Takes over exactly where the muonic regime ends, targeting interstellar missions.

Connection to this group's themes. The same geometric lepton structure proposed here to explain g-2 may be relevant to the physics of jets from compact objects: if massive leptons interact with the quantum vacuum at amplified rates proportional to *$(m_\ell/m_e)^5$*, environments producing high populations of muons or tauons — such as the extreme conditions near black hole ergospheres or in relativistic jet formation zones — could exhibit vacuum coupling effects not captured by standard QED. The three-generation mass hierarchy of leptons may encode a natural ladder of vacuum

interaction scales with potential implications for jet energy budgets and neutrino production mechanisms.

Recent experimental progress on compact muon sources (CAEP/CAS laser-driven muon production, *Nature Physics* 2025; J-PARC muon acceleration to 100 keV; MICE ionization cooling) makes this framework timely.

The full text, including all derivations and numerical calculations, is available upon request.

Comments, critiques, and questions from the community are most welcome.

Best regards,

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